

IT Help Desk & Ticketing Management System

Full Stack Web Development Internship Project



INTEGRATED DIGITAL
SYSTEMS



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1. Project Overview

The purpose of this project is to design and develop a modern web-based IT Help Desk & Ticketing Management System that will help streamline technical support operations inside the company.

Employees will be able to submit support requests, while IT support agents and administrators can manage, prioritize, assign, and resolve tickets through a centralized dashboard.

The project will simulate a real-world enterprise software development environment and provide interns with hands-on experience in frontend development, backend APIs, database design, authentication, reporting, deployment, and AI integrations.

2. Project Objectives

By the end of the internship, interns should be able to:

- Build a complete enterprise-level full stack web application
- Design and implement RESTful APIs
- Create responsive and modern user interfaces
- Manage relational databases
- Implement authentication and role-based authorization
- Build dashboards and reporting systems
- Work using Git and collaborative workflows
- Deploy applications to production/staging environments
- Integrate AI-powered automation features

a) Suggested Technology Stack

Frontend

- React.js / Next.js
- Tailwind CSS
- Shadcn UI / Material UI

Backend

- ASP.NET Core Web API
- Node.js Express (optional alternative)

Database

- SQL Server / PostgreSQL

Authentication

- JWT Authentication
- ASP.NET Identity

Deployment

- IIS / Azure / Docker

AI Integration

- OpenAI API / Azure OpenAI
 - Ollama (Optional local AI)
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b) System Users & Roles

Role	Permissions
Admin	Full system access
IT Support Agent	Manage and resolve tickets
Employee	Create and track tickets
Manager	Monitor team tickets and reports

3. System Modules & Features

a) Authentication & User Management

Features

- User registration/login
- Password encryption
- Forgot/reset password
- Profile management
- Role-based access control
- Session/token management

Requirements

- Secure authentication using JWT
- Password validation
- Protected API routes

- User activity logging

b) Ticket Management Module

Features

- Create support tickets
- Edit/update tickets
- Delete/cancel tickets
- Ticket history tracking
- Ticket categories:
 - Hardware
 - Software
 - Network
 - Email
 - Access Request
 - Other

Ticket Priorities

- Low
- Medium
- High
- Critical

Ticket Statuses

- Open
- In Progress
- Pending
- Resolved
- Closed

Requirements

- Validation for required fields
- Ticket reference number generation
- Search and filtering functionality

c) Ticket Assignment & Workflow

Features

- Assign tickets to support agents
- Manual or automatic assignment
- Escalation workflow
- Reassign tickets
- Internal comments/notes

Requirements

- Only authorized roles can assign tickets
- Track assignment history
- Audit trail for ticket actions

d) Communication & Notifications

Features

- In-app notifications
- Email notifications
- Ticket update alerts
- Comment/reply system
- Mention/tag support agents

Requirements

- Real-time updates (optional using SignalR/WebSockets)
- Notification center dashboard

e) Dashboard & Reporting

Dashboard Widgets

- Open tickets count
- Resolved tickets count
- Pending tickets
- Tickets by category
- Tickets by priority
- Agent performance charts

Reports

- Monthly ticket reports
- Average resolution time
- SLA reports (optional)
- Employee activity reports

Requirements

- Export reports to PDF/Excel
- Charts and analytics

f) Knowledge Base Module (Optional Advanced Module)

Features

- FAQ management
- Troubleshooting articles
- Searchable documentation
- Category-based articles

Requirements

- Rich text editor
- Search functionality
- Admin approval for articles

g) File Attachments

Features

- Upload screenshots/documents
- Attach logs or files to tickets
- Download attachments securely

Requirements

- File size validation
- Supported file types validation
- Secure file storage

h) Admin Panel

Features

- User management

- Role management
- Ticket categories management
- System settings
- Reports generation

Requirements

- Restricted admin access
- Activity logging
- System monitoring dashboard

i) AI-Powered Features (Advanced)

AI Ticket Categorization

Automatically classify tickets based on description.

Example:

“Outlook not working”

→ Software Issue

AI Priority Suggestion

AI recommends ticket urgency.

Example:

“Main server is offline”

→ Critical Priority

AI Suggested Replies

Generate troubleshooting suggestions for support agents.

AI Chat Assistant

Employees can ask questions before creating tickets.

Example:

“How do I connect to VPN?”

→ AI returns setup instructions.

4. UI/UX Requirements

Interns should design a modern and responsive SaaS-style interface.

Suggested Pages

- Login/Register
- Dashboard
- Ticket List
- Ticket Details
- Create Ticket
- Reports
- Notifications
- User Profile
- Admin Settings

Design Requirements

- Responsive design
- Mobile-friendly layout
- Sidebar navigation
- Dark/light mode (optional)
- Loading states & error handling

Database Requirements

Interns must design a relational database including:

Suggested Tables

- Users
- Roles
- Tickets
- TicketComments
- TicketAttachments
- Notifications
- Categories
- Priorities
- Statuses
- ActivityLogs

Deliverables

- ERD Diagram
- SQL scripts
- Seed/sample data

5. Weekly Timeline & Assignments (8 Weeks)

Week	Tasks	Deliverables	Estimated Hours
1	Requirement analysis, project planning, UI wireframes, ERD/database schema	Wireframes, ERD, project proposal	25 hrs
2	Project setup, authentication, role management	Login/Register system, JWT auth	25 hrs
3	Ticket CRUD operations, ticket categories & priorities	Ticket module functional	30 hrs
4	Assignment workflow, comments, ticket statuses	Workflow implementation	30 hrs
5	Notifications, file uploads, dashboard analytics	Dashboard & notification system	30 hrs
6	Reports, charts, export functionality, AI integration basics	Reporting module & AI prototype	30 hrs
7	Testing, bug fixing, optimization, responsive UI improvements	Stable staging version	30 hrs
8	Deployment, documentation, final presentation/demo	Final deployed project	40 hrs

Weekly Submission Requirements

Each week interns must submit:

- Updated source code
- GitHub commits/repository
- Weekly progress report
- Screenshots or demo videos
- Completed assigned tasks
- Updated documentation

Weekly review meetings will be conducted to evaluate progress and provide technical guidance.

6. Final Deliverables

At the end of the internship, interns must provide:

- Full source code
- GitHub repository
- README documentation
- Database schema (ERD + SQL scripts)
- API documentation
- Setup instructions
- Screenshots/demo videos
- Deployment link (if available)
- Final presentation/demo

Optional Bonus Features

High-performing interns may implement:

- Real-time chat support
- SLA timer system
- Microsoft Teams integration
- Email-to-ticket automation
- QR code asset management
- Audit logs
- Multi-language support
- Dark mode
- Mobile app version
- Docker deployment
- CI/CD pipeline

7. Recommended Development Workflow

Interns are encouraged to use:

- Git & GitHub
- Branching strategy
- Pull requests
- Agile task management
- Weekly sprint reviews

Expected Learning Outcomes

By the end of the internship, interns will gain practical experience in:

- Full stack web development
- API development
- Database architecture
- Authentication & security
- Team collaboration
- Enterprise workflows
- AI-assisted applications
- Software deployment & maintenance